DINITH SHEHAN JIFFRY

C++ Software Engineer | Financial Infrastructure & Exchange Systems

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PROFESSIONAL SUMMARY

Software Engineer with 5+ years working for the London Stock Exchange Group (LSEG) developing back-end financial infrastructure and clearing systems. Expertise in C++11, Linux system development, Apache Kafka connectivity, and low-level debugging using GDB and Valgrind. Familiar with conducting sprint ceremonies as a Scrum Master and developing developer tools to automate environment setup.

TECHNICAL SKILLS

- **Languages:** C++, Python, Java, SQL (PostgreSQL, Oracle).
- **Systems & Infrastructure:** Multithreading, Memory Management, Distributed Systems, Unix/Linux, Apache Kafka.
- **Debugging Tools:** GDB, Valgrind, Google Benchmark, Coverity, Git, Bash.
- **Domain:** Central Counterparty (CCP) Clearing, Market Infrastructure, Automated Testing, Agile/Scrum.

PROFESSIONAL EXPERIENCE

London Stock Exchange Group | Senior Software Engineer (CCP Team) | Mar 2024 – Present

- **A5X Core Platform Architecture:** Developed core C++ modules for exchange connectivity, including trade-leg correlation messaging pipelines, real-time MTM obligation enrichment, posting validations, and collateral response processing capabilities.
- **Data Structure and Latency Improvements:** Incorporated caching in memory to avoid high-latency database queries for account linking. Profiling through application log analysis revealed a significant drop in lookup latency, eliminating the I/O bottlenecks that used to be encountered during volume spikes.
- **Kafka Messaging Library:** Implemented a thread-safe, TLS-encrypted C++ communication library for CSV/JSON encoding. Profiling using Valgrind was done to remove dynamic memory allocation on key paths and guarantee no memory leaks or concurrency problems at high loads.
- **System Stability and Archival:** Developed transactional archiving and EOD purging functionality to get rid of database bloat issues that occurred during heavy settlement processing. Proved through system performance metrics where we saw no more database spikes after settlement.
- **Release Automation of AI:** Created an AI agent through custom-built Markdown workflows to enable automated sprint-level dependency management for 8 repositories (4 core dependencies and 4 solutions). Achieved automated branch creation, custom code migration, and conflict resolution, reducing release upgrade time from 3 days to 1 day per sprint.
- **Developer Tooling and Environments:** Automated topic, partitions, and mock server

creation using Bash scripts and Kafka utilities, as well as an SQL stored procedure for automated system gateway creation.
- **Debugging and Code Quality:** Debugged and addressed edge-case memory leaks with GDB and Valgrind in multi-threaded applications.
- **Quality Governance & Defect Triage:** Transitioned Coverity static analysis from being used as a post hoc compliance check to a requirement before release; rigorously triaged scan results, eliminating false positives, and fixed critical memory and logic defects such that number of open issues is kept below thresholds.

London Stock Exchange Group | **Software Engineer (CCP Team)** | **Oct 2021 – Mar 2024**
- **Qatar Exchange Migration:** Adapters created for LSEG's clearing engine to map Qatar Exchange trade processing logic and market rules into LSEG's core CCP process flows. Created real-time messaging parsing/validations processes to ensure loss-less trade intake during high volume migration testing.
- **European Clearing Services:** Resolved key C++ bugs in production including race condition in multithreading and memory leak in special cases for Paris RCP and London RCL clearing systems.
- **Testing Framework:** Created unit tests, integration tests, and regression tests for the system in C++, taking care of edge cases under heavy load.
- **Scrum Master:** Managed Scrum Master tasks after a transition of senior team members, facilitating agile development for one year before taking on the role of leading technical triage for important CCP/Exchange integration testing.

London Stock Exchange Group | **Software Engineer Intern** | **Feb 2019 – July 2019**
- **Benchmarking:** Measured STL container performance through Google Benchmark for the selection of containers used in the core engine.
- **Code Refactoring:** Refactored old C++ code bases by getting rid of warnings in the compiler.
- **SQL Database Migration:** Validated and migrated data sets from Oracle SQL to PostgreSQL.

EDUCATION

Sri Lanka Institute of Information Technology (SLIIT) | **Jan 2017 – August 2021**

- **BSc (Hons) in Information Technology (Software Engineering)**
- **GPA: 3.29/4.00**
- **Coursework:** Data Structures & Algorithms, Distributed Systems, Parallel Computing, OS Administration, Secure Software Development.

SYSTEMS PROJECTS

- **Turtlebot Maze Navigation (C++/ROS):** Developed a path finding algorithm utilizing ROS that enables the robot to navigate autonomously.
- **Fire Detection System (Java/API):** Constructed a multi-instance safety system capable of recovering from independent service failure.
- **Resource Security Access Control (C++/Java):** Developed an access control system with Apache Felix API for modular resource access.

CERTIFICATIONS & ACHIEVEMENTS

- **IELTS Academic (2021): Overall Band Score: 9.0** (Listening: 9.0, Reading: 8.0, Speaking: 8.0, Writing: 7.0).
- **Professional Development:** Certificates in Public Speaking and Personal/Professional Development (American College in Higher Education).